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Edge band fine trimming mechanism of edge banding machine

Abstract

An edge band fine trimming mechanism of an edge banding machine includes a fixed seat including a scraper installation groove extending vertically, a first feeler wheel assembly including a first feeler wheel vertically and rotatably disposed on the right side of the fixed seat, a second feeler wheel assembly including a second feeler wheel horizontally and rotatably disposed on the bottom side of the fixed seat, and a scraper assembly. The aforementioned scraper assembly is disposed on the fixed seat and includes a blade holder and a blade. The blade holder includes vertical and inclined sections connected with each other. The vertical section is disposed in the scraper installation groove in a vertically movable manner. The blade is disposed at the terminal end of the inclined section. The edge band fine trimming mechanism enables fast positional adjustment for the scraper and the associated feeler wheel.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an edge band fine trimming mechanism of an edge banding machine and specifically speaking, to an edge band fine trimming mechanism which enables relatively easier positional control for a scraper.

2. Description of the Related Art

(2) As to the traditional edge banding machine, refer to FIG. **1**, wherein an edge banding machine **1** is illustrated, which includes a base **10**, a set of slideway **20**, a positioning pressing wheel set (not

shown), and a plurality of other mechanisms including a pre-milling mechanism **30**, a glue applying mechanism **40**, an edge band conveying mechanism **50**, a horizontal pressing wheel set **60** and an edge band fine trimming mechanism **70**. The slideway **20** is movably disposed on the base **10**. The positioning pressing wheel set is disposed above the slideway **20**. A workpiece (not shown) to be processed is clipped between the slideway **20** and the positioning pressing wheel set, and can be moved. The pre-milling mechanism **30** performs a preliminary processing to the surface of the workpiece. The glue applying mechanism **40** applies glue to the surface of the workpiece. The edge band conveying mechanism **50** is adapted to convey an edge band to the workpiece applied with the glue. Then, the edge banding machine **1** cuts off the edge band by a cutter **51** and tightly presses the cut-off edge band to the workpiece by the horizontal pressing wheel set **60**, so as to combine the cut-off edge band with the workpiece applied with the glue tightly. At last, the edge banding machine **1** further trims end edges of the edge band by the edge band fine trimming mechanism **70**, so as to chamfer the end edges of the edge band.

(3) In the edge trimming process by using the edge band fine trimming mechanism **70**, it needs feeler wheels to keep a scraper of the edge band fine trimming mechanism **70** located at a fixed position relative to the workpiece, preventing the edge trimming process from causing an end edge of the edge band an appearance defect. According to what the inventor knows, the traditional scraper is structurally shaped as a linear rod, and arranged at an angle of 45 degrees with respect to the ground. Besides, the traditional scraper is adjusted along the linear extending direction of the scraper. Therefore, once the position of the scraper is adjusted, it will certainly change two of the positional coordinates of the blade tip of the scraper at the same time, resulting in that the technician for the edge banding machine **1** has to further correspondingly correct the positions of the feeler wheels in the following process, thereby causing inconvenience in correction. It is thus clear that the structural design of the edge band fine trimming mechanism **70** of the presently available edge banding machine **1** is imperfect and thereby still needs improvement.

SUMMARY OF THE INVENTION

(4) It is one of the objectives of the present invention to make an improvement against the defects of the edge band fine trimming mechanism of the presently available edge banding machine, so as to bring up a brand-new configuration design of an edge band fine trimming mechanism of an edge banding machine, which enables fast and convenient positional adjustment for the scraper and the associated feeler wheel, lowering the inconvenience in correcting the scraper and the feeler wheel.

(5) Therefore, an edge band fine trimming mechanism of an edge banding machine provided according to the present invention is applied to a workpiece with an edge band adhered thereto. The aforementioned edge band fine trimming mechanism includes a fixed seat, a first feeler wheel assembly, a second feeler wheel assembly, and a scraper assembly. The aforementioned fixed seat includes a scraper installation groove, and the scraper installation groove extends vertically. The aforementioned first feeler wheel assembly includes a first feeler wheel. The first feeler wheel is vertically and rotatably disposed on the front side of the fixed seat. The aforementioned second feeler wheel assembly includes a second feeler wheel. The second feeler wheel is horizontally and rotatably disposed on the bottom side of the fixed seat. The aforementioned scraper assembly is disposed on the fixed seat and adapted to trim an end edge of the edge band. The scraper assembly includes a blade holder and a blade. The blade holder includes a vertical section and an inclined section. The vertical section is disposed in the scraper installation groove in a vertically movable manner. The inclined section is connected to the bottom end of the vertical section and extends toward a boundary between the aforementioned first feeler wheel and the second feeler wheel. The blade is disposed at the terminal end of the inclined section.

(6) As a result, when the edge banding process is to be performed to a workpiece of different size and thereby the position of the scraper assembly needs adjustment, the technician for the edge banding machine only needs to vertically adjust the scraper assembly along the scraper installation groove. During the adjusting process, only one of the coordinates of the blade tip of the scraper

assembly is changed, so that the technician for the edge banding machine only needs to correspondingly adjust the position of the first feeler wheel in the following process to complete the entire correcting process for the edge band fine trimming mechanism. Therefore, the positions of the scraper and the associated feeler wheel can be adjusted fast and conveniently, that effectively lowers the inconvenience in correcting the scraper and the feeler wheel.
(7) In one of the aspects, the aforementioned edge band fine trimming mechanism may be a top trimming mechanism or a bottom trimming mechanism for trimming a top end edge or a bottom end edge of the edge band respectively.
(8) In another aspect, the fixed seat further has an upper threaded hole. The scraper assembly further includes a first adjusting screw. The first adjusting screw is screwed into the upper threaded hole of the fixed seat, and the terminal end of the first adjusting screw is connected with the blade holder.
(9) Specifically, the scraper assembly further includes a first displacement sensor. The first displacement sensor is disposed on the first adjusting screw and measures the vertical displacement of the first adjusting screw.
(10) In another aspect, the edge band fine trimming mechanism further includes a supporting frame and a second adjusting screw. The supporting frame is disposed on the fixed seat in a horizontally movable manner. The second feeler wheel is rotatably disposed at the terminal end of the supporting frame. The fixed seat further has a lower threaded hole. The second adjusting screw is screwed into the lower threaded hole of the fixed seat, and the terminal end of the second adjusting screw is connected with the supporting frame.
(11) Specifically, the second feeler wheel assembly further includes a second displacement sensor. The second displacement sensor is disposed on the second adjusting screw and measures the horizontal displacement of the second adjusting screw.
(12) In another aspect, the fixed seat has a fastening threaded hole. The vertical section has an elongated hole. A fastening screw is inserted through the elongated hole and screwed into the fastening threaded hole for adjusting the position of the scraper assembly.
(13) In another aspect, the first feeler wheel assembly includes a plurality of first feeler wheels and a fixing member. The fixing member is disposed on the fixed seat in a vertically movable manner. The plurality of first feeler wheels are all vertically and rotatably disposed on the fixing member. The plurality of first feeler wheels are all located at a same predetermined height.
(14) In another aspect, the scraper installation groove has a first inner wall and a second inner wall, which are arranged oppositely to each other. The peripheral surface of the blade holder fitly contacts the first inner wall and the second inner wall at the same time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The detailed structure, features, assembling or using manner related to the edge band fine trimming mechanism of the edge banding machine will be described in the following embodiment. However, it should be understood that the embodiment to be described in the following and the figures are given by way of illustration only, not limitative of the claimed scope of the present invention, and wherein:
(2) FIG. **1** is a perspective view of a conventional edge banding machine;
(3) FIG. **2** is a perspective view of an edge band fine trimming mechanism of an edge banding machine of an embodiment;
(4) FIG. **3** is a lateral view of FIG. **2**;
(5) FIG. **4** is a bottom view of FIG. **2**; and
(6) FIG. **5** is a schematic view showing the usage of the edge band fine trimming mechanism of the

embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(7) The technical content and features of the present invention will be described in detail by the following embodiment in coordination with the figures. The directional terms mentioned in the content of the specification, such as ‘upper’, ‘lower’, ‘in’, ‘out’, ‘top’, and ‘bottom’, are just for illustrative description on the basis of normal usage direction, not intended to limit the claimed scope.

(8) For the detailed description of the technical features of the present invention, the following embodiment is instanced and illustrated in coordination with the figures. Wherein, for the convenience of illustrating the following embodiment, the direction of this embodiment will be based on that the workpiece W is located on the right side of the edge band fine trimming mechanism **10**.

(9) As shown in FIG. **2**, an edge band fine trimming mechanism **10** of an edge banding machine provided in the embodiment is applied to a workpiece W (shown in FIG. **5**) with an edge band B adhered thereto for chamfering top, bottom, front and rear end edges of the edge band B. The workpiece W is disposed on a set of slideway R, and a set of positioning pressing wheel set P is abutted on the top surface of the workpiece W. The workpiece W is moved by the set of slideway R and the set of positioning pressing wheel set P.

(10) Referring to FIG. **2** to FIG. **5**, the aforementioned edge band fine trimming mechanism **10** includes a fixed seat **20**, a first feeler wheel assembly **30**, a second feeler wheel assembly **40**, and a scraper assembly **50**.

(11) The fixed seat **20** is adapted for disposing the first feeler wheel assembly **30**, the second feeler wheel assembly **40** and the scraper assembly **50**, and the fixed seat **20** is fixed on a base of the edge banding machine. The fixed seat **20** is provided on the right side thereof with two guiding rods **21** extending vertically, and the fixed seat **20** is provided on the front side thereof with a scraper installation groove **22**. The scraper installation groove **22** extends vertically and has a first inner wall **221** and a second inner wall **222**, which are arranged oppositely to each other. The so-called ‘extend vertically’ means extending upwardly with respect to the ground.

(12) The first feeler wheel assembly **30** includes five first feeler wheels **31** and a fixing member **32**. The first feeler wheels **31** are vertically and rotatably disposed on the fixing member **32** and located on the right side of the fixed seat **20**. The fixing member **32** is disposed on the two guiding rods **21** in a vertically movable manner. The so-called ‘vertically disposed on the fixing member **32**’ means the axial directions of the axles of the first feeler wheels **31** are all parallel to the ground. The bottom ends of the first feeler wheels **31** all a little protrude out below the fixing member **32**, and the first feeler wheels **31** are all located at a same predetermined height. The first feeler wheels **31** are in contact with the top surface of the workpiece W for determining the position of a top reference surface of the workpiece W.

(13) The second feeler wheel assembly **40** includes a second feeler wheel **41**, a supporting frame **42**, a second adjusting screw **43**, and a second displacement sensor **44**. The second feeler wheel **41** is horizontally and rotatably disposed on the bottom side of the fixed seat **20**. The second feeler wheel **41** is in contact with the surface of the edge band B for determining the position of a lateral reference surface of the workpiece W with the edge band B adhered thereto. As shown in FIG. **3** and FIG. **4**, the so-called ‘horizontally disposed on the fixed seat **20**’ means the axial direction of the axle of the second feeler wheel **41** is vertical to the ground. The supporting frame **42** is disposed on the fixed seat **20** in a horizontally movable manner. The supporting frame **42** has a first section **421** and a second section **422**, which are connected with each other. The first section **421** extends from the front to the rear. The second section **422** extends from the right to the left. The direction which the first section **421** extends toward is perpendicular to the direction which the second section **422** extends toward. The second feeler wheel **41** is rotatably disposed on the terminal end of the first section **421** of the supporting frame **42**. The fixed seat **20** further has a

lower threaded hole **24**. The second adjusting screw **43** is screwed into the lower threaded hole **24** of the fixed seat **20**, and a terminal end of the second adjusting screw **43** is connected to the second section **422** of the supporting frame **42**. The other end of the second adjusting screw **43** is provided with a handle **45**. The second displacement sensor **44** is disposed on the second adjusting screw **43** and measures the horizontal displacement of the second adjusting screw **43**. Rotating the handle **45** can drive the supporting frame **42** to displace horizontally, thereby adjusting the position of the second feeler wheel **41**. For effectively fixing the supporting frame **42**, the second section **422** of the supporting frame **42** is provided with an elongated hole **423**, and the bottom side of the fixed seat **20** is provided with a threaded hole (covered by a screw S and thereby not shown). The threaded hole is located correspondingly to the elongated hole **423**. The screw S is inserted through the elongated hole **423** and screwed into the aforementioned threaded hole, so that the supporting frame **42** can be fixed to the fixed seat **20** by the screw S after being adjusted in position.

(14) Referring to FIG. **3** and FIG. **5**, the scraper assembly **50** is disposed on the fixed seat **20** and adapted to trim an end edge of the edge band B. The aforementioned end edge of the edge band B may be a top end edge, a bottom end edge, a front end edge or a rear end edge of the edge band B. The scraper assembly **50** structurally includes a blade holder **51**, a blade **52**, a first adjusting screw **53**, and a first displacement sensor **54**. The blade holder **51** includes a vertical section **511** and an inclined section **512**. The vertical section **511** is disposed in the scraper installation groove **22** in a vertically movable manner, and the peripheral surface of the blade holder **51** fitly contacts the first inner wall **221** and the second inner wall **222** of the scraper installation groove **22** at the same time. The inclined section **512** is connected to the bottom end of the vertical section **511** and extends toward a boundary between the first feeler wheels **31** and the second feeler wheel **41**. The blade **52** is disposed at the terminal end of the inclined section **512**. For effectively fixing the blade holder **51** when the blade **52** and the blade holder **51** are adjusted together to the position corresponding to the top end edge E of the edge band B, the vertical section **511** of the blade holder **51** is provided with an elongated hole **513** (as shown in FIG. **5**), and the fixed seat **20** is provided with a fastening threaded hole **23** (located in the elongated hole **513** in FIG. **5**). A fastening screw (not shown) is inserted through the elongated hole **513** and screwed into the aforementioned fastening threaded hole **23**, so that the blade holder **51** can be fixed in the scraper installation groove **22** of the fixed seat **20**. The first adjusting screw **53** is screwed into an upper threaded hole **25** of the fixed seat **20**, and a terminal end of the first adjusting screw **53** is connected with the blade holder **51**. The first displacement sensor **54** is disposed on the first adjusting screw **53** and measures the vertical displacement of the first adjusting screw **53**.

(15) Referring to FIG. **5**, by the configuration design of the edge band fine trimming mechanism **10** of this embodiment, when the edge banding process is to be performed to a workpiece W of different size and thereby the position of the scraper assembly **50** needs adjustment, the technician for the edge banding machine only needs to rotate the first adjusting screw **53** to vertically adjust the blade holder **51** and the blade **52** of the scraper assembly **50** together along the scraper installation groove **22**. During the adjusting process, only one of the coordinates of the blade tip of the blade **52** is changed, which means the blade **52** is only moved vertically, so that the technician for the edge banding machine only needs to correspondingly adjust the position of the first feeler wheels **31** in the following process to complete the entire correcting process for the edge band fine trimming mechanism **10**. Therefore, the positions of the scraper assembly **50** and the second feeler wheel **41** can be adjusted fast and conveniently, that effectively lowers the inconvenience in correcting the scraper assembly **50** and the second feeler wheel **41**.

(16) The invention being thus described, it will be obvious that the same may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the invention, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

Claims

1. An edge band fine trimming mechanism of an edge banding machine, which is applied to a workpiece with an edge band adhered thereto, the edge band fine trimming mechanism comprising: a fixed seat comprising a scraper installation groove, the scraper installation groove extending vertically; a first feeler wheel assembly comprising a first feeler wheel, the first feeler wheel being vertically and rotatably disposed on a right side of the fixed seat; a second feeler wheel assembly comprising a second feeler wheel, the second feeler wheel being horizontally and rotatably disposed on a bottom side of the fixed seat; a scraper assembly disposed on the fixed seat and adapted to trim an end edge of the edge band, the scraper assembly comprising a blade holder and a blade, the blade holder comprising a vertical section and an inclined section, the vertical section being vertically and movably disposed in the scraper installation groove, the inclined section being connected to a bottom end of the vertical section and extending toward a boundary between the first feeler wheel and the second feeler wheel, the blade being disposed at a terminal end of the inclined section; wherein the fixed seat further has an upper threaded hole; the scraper assembly further comprises a first adjusting screw; the first adjusting screw is screwed into the upper threaded hole of the fixed seat and a terminal end of the first adjusting screw is connected with the blade holder.
2. The edge band fine trimming mechanism as claimed in claim 1, wherein the scraper assembly further comprises a first displacement sensor; the first displacement sensor is disposed on the first adjusting screw and measures a vertical displacement of the first adjusting screw.
3. The edge band fine trimming mechanism as claimed in claim 1, wherein the fixed seat has a fastening threaded hole; the vertical section has an elongated hole; a fastening screw is inserted through the elongated hole and screwed into the fastening threaded hole.
4. The edge band fine trimming mechanism as claimed in claim 1, wherein the scraper installation groove has a first inner wall and a second inner wall, which are arranged oppositely to each other; a peripheral surface of the blade holder fitly contacts the first inner wall and the second inner wall at a same time.
5. An edge band fine trimming mechanism of an edge banding machine, which is applied to a workpiece with an edge band adhered thereto, the edge band fine trimming mechanism comprising: a fixed seat comprising a scraper installation groove, the scraper installation groove extending vertically; a first feeler wheel assembly comprising a first feeler wheel, the first feeler wheel being vertically and rotatably disposed on a right side of the fixed seat; a second feeler wheel assembly comprising a second feeler wheel, the second feeler wheel being horizontally and rotatably disposed on a bottom side of the fixed seat; a scraper assembly disposed on the fixed seat and adapted to trim an end edge of the edge band, the scraper assembly comprising a blade holder and a blade, the blade holder comprising a vertical section and an inclined section, the vertical section being vertically and movably disposed in the scraper installation groove, the inclined section being connected to a bottom end of the vertical section and extending toward a boundary between the first feeler wheel and the second feeler wheel, the blade being disposed at a terminal end of the inclined section; wherein the second feeler wheel assembly further comprises a supporting frame and a second adjusting screw; the supporting frame is disposed on the fixed seat in a horizontally movable manner; the second feeler wheel is rotatably disposed at a terminal end of the supporting frame; the fixed seat further has a lower threaded hole; the second adjusting screw is screwed into the lower threaded hole of the fixed seat and a terminal end of the second adjusting screw is connected with the supporting frame.
6. The edge band fine trimming mechanism as claimed in claim 5, wherein the second feeler wheel assembly further comprises a second displacement sensor; the second displacement sensor is disposed on the second adjusting screw and measures a horizontal displacement of the second

adjusting screw.

7. The edge band fine trimming mechanism as claimed in claim 5, wherein the supporting frame has a first section and a second section; the first section is connected with the second section; a direction which the first section extends toward is perpendicular to another direction which the second section extends toward; the second feeler wheel is rotatably disposed on the first section; the second adjusting screw is connected with the second section.

8. The edge band fine trimming mechanism as claimed in claim 7, wherein the second section further has an elongated hole; the bottom side of the fixed seat has a threaded hole; the threaded hole is located correspondingly to the elongated hole; a screw is inserted through the elongated hole of the second section and screwed into the threaded hole.

9. An edge band fine trimming mechanism of an edge banding machine, which is applied to a workpiece with an edge band adhered thereto, the edge band fine trimming mechanism comprising: a fixed seat comprising a scraper installation groove, the scraper installation groove extending vertically; a first feeler wheel assembly comprising a first feeler wheel, the first feeler wheel being vertically and rotatably disposed on a right side of the fixed seat; a second feeler wheel assembly comprising a second feeler wheel, the second feeler wheel being horizontally and rotatably disposed on a bottom side of the fixed seat; a scraper assembly disposed on the fixed seat and adapted to trim an end edge of the edge band, the scraper assembly comprising a blade holder and a blade, the blade holder comprising a vertical section and an inclined section, the vertical section being vertically and movably disposed in the scraper installation groove, the inclined section being connected to a bottom end of the vertical section and extending toward a boundary between the first feeler wheel and the second feeler wheel, the blade being disposed at a terminal end of the inclined section; wherein the first feeler wheel assembly comprises a plurality of said first feeler wheels and a fixing member; the fixing member is disposed on the fixed seat in a vertically movable manner; the plurality of said first feeler wheels are all vertically and rotatably disposed on the fixing member; the plurality of said first feeler wheels are all located at a same predetermined height.